EXPLORATORY DATA ANALYSIS

Forbes Richest Athletes

1990 – 2020

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1. Introduction

This exploratory data analysis (EDA) examines the earnings of top-ranked athletes across different sports, countries, and time periods (1990–2020). The purpose of the analysis is to understand earnings distributions, identify trends over time, and explore the relationship between athlete rankings and earnings. The analysis involves data cleaning, descriptive statistics, data visualisation, and correlation analysis to uncover patterns and potential anomalies. The insights gained provide a foundation for further statistical analysis and support data-driven conclusions about the factors influencing athlete earnings.

2. Dataset Overview

Total Records	Unique Athletes	Unique Sports	Nationalities
301	**82**	**20**	**22**

The dataset contains 301 records spanning 31 years (1990–2020), covering 82 unique athletes across 20 sports and 22 nationalities. Many athletes appear multiple times, reflecting repeat appearances on the Forbes list over successive years.

Sample Data Structure

Name	Nationality	Current Rank	Sport	Year	Earnings ($M)
Mike Tyson	USA	1	Boxing	1990	28.6
Buster Douglas	USA	2	Boxing	1990	26.0
Ayrton Senna	Brazil	4	Auto Racing	1990	10.0
Alain Prost	France	5	Auto Racing	1990	9.0
Michael Jordan	USA	8	Basketball	1990	8.1

3. Data Cleaning & Missing Values

The data cleaning process focused on ensuring accuracy, consistency, and usability of the dataset prior to analysis. Initial inspection was performed to identify missing values, inconsistent formats, and potential anomalies.

- **Missing values:** Only the *Previous Year Rank* column contained missing data — 24 values (7.97%). These were retained as NaN because they are meaningful: they indicate the athlete was not ranked in the prior year (new entrant or returning athlete).
- **Column standardisation:** Column names were converted to lowercase with underscores for consistency.
- **Sport names:** Converted to title case to eliminate inconsistencies (e.g. 'boxing' → 'Boxing').
- **Previous Year Rank:** Values recorded as '>30' were replaced with NaN, as the exact rank was unknown.

• **Numeric types:** Earnings and previous_year_rank columns were cast to appropriate numeric types.

• **Duplicates:** Removed to prevent double-counting of observations.

Column	Missing Values	Percentage Missing
name	0	0.00%
nationality	0	0.00%
current_rank	0	0.00%
previous_year_rank	24	7.97%
sport	0	0.00%
year	0	0.00%
earnings	0	0.00%

4. Descriptive Statistics

Mean Earnings	Median Earnings	Std. Deviation	Maximum
$45.52M	**$39.00M**	**$33.53M**	**$300.00M**

Statistic	Current Rank	Prev. Year Rank	Year	Earnings ($M)
Count	301	212	301	301
Mean	5.45	7.17	2005.12	45.52
Std	2.85	6.40	9.06	33.53
Min	1.00	1.00	1990	8.10
25%	3.00	3.00	1997	24.00
50%	5.00	6.00	2005	39.00
75%	8.00	9.00	2013	59.40
Max	10.00	40.00	2020	300.00

Earnings display a right-skewed distribution. The mean ($45.52M) sits considerably above the median ($39M), driven by a small number of extreme outliers. The standard deviation of $33.53M reflects large disparity between typical top-earners and elite superstars such as Floyd Mayweather.

5. Data Visualisation

5.1 Earnings Distribution

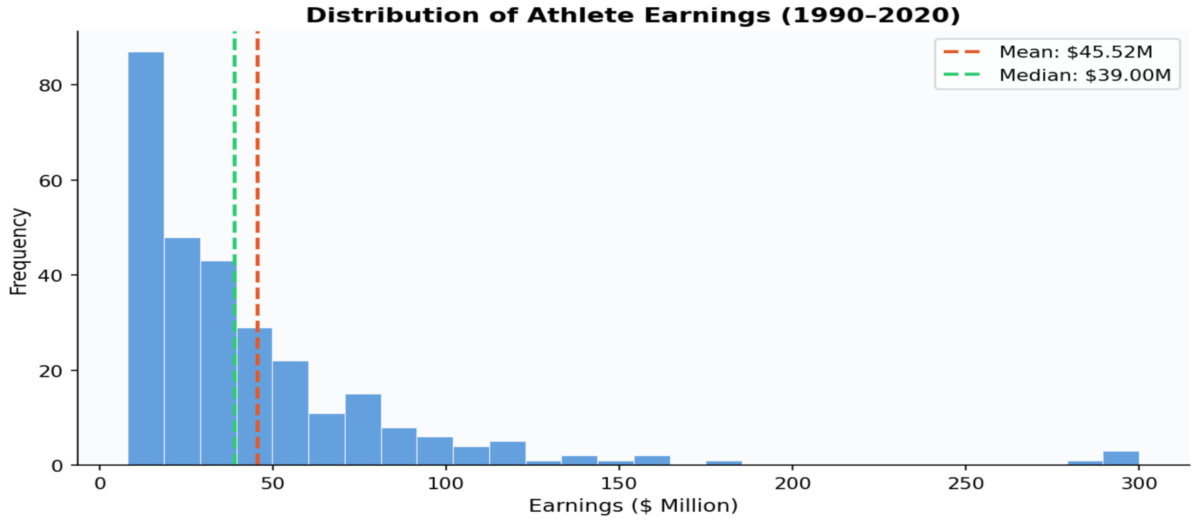

Figure 1 – Distribution of athlete earnings (1990–2020)

→ 67.4% of athletes earn between $0–50M (the peak of the distribution).

→ The distribution has a long tail extending to $300M (Floyd Mayweather, 2015 — highest single-year earner).

→ Only 2 observations exceed $200M, both attributed to Floyd Mayweather.

→ The mean ($45.52M) is significantly higher than the median ($39M) due to extreme outliers — a classic **superstar effect**.

5.2 Sport Distribution

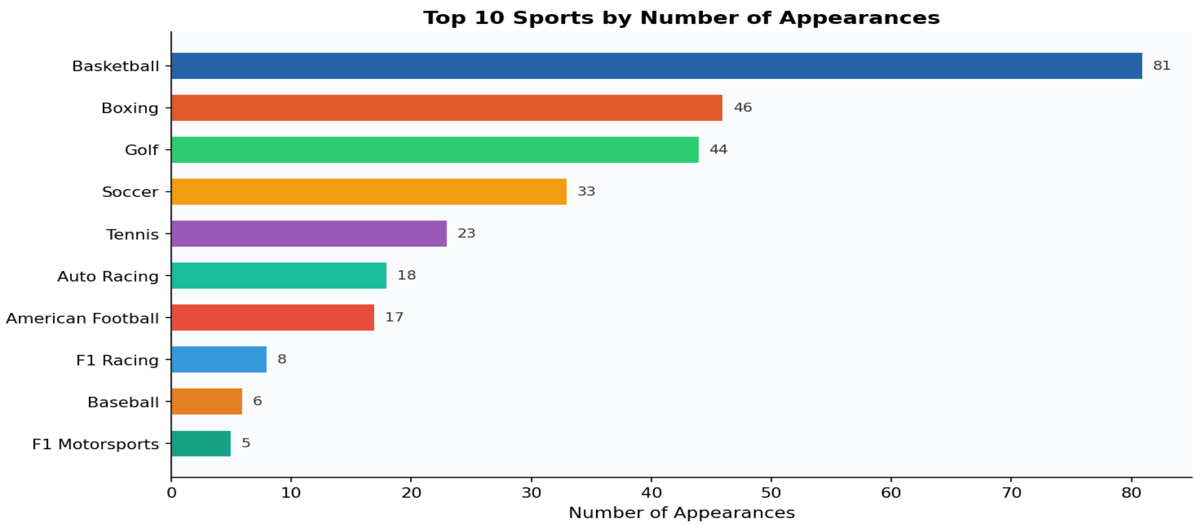

Figure 2 – Top 10 sports by number of list appearances

→ Basketball dominates with 81 appearances — roughly 2× more than Boxing (46).

→ The top 3 sports (Basketball, Boxing, Golf) account for **56.8%** of all appearances.

→ Individual sports (Boxing, Golf, Tennis) show wider earnings variability than team sports.

5.3 Nationality Distribution

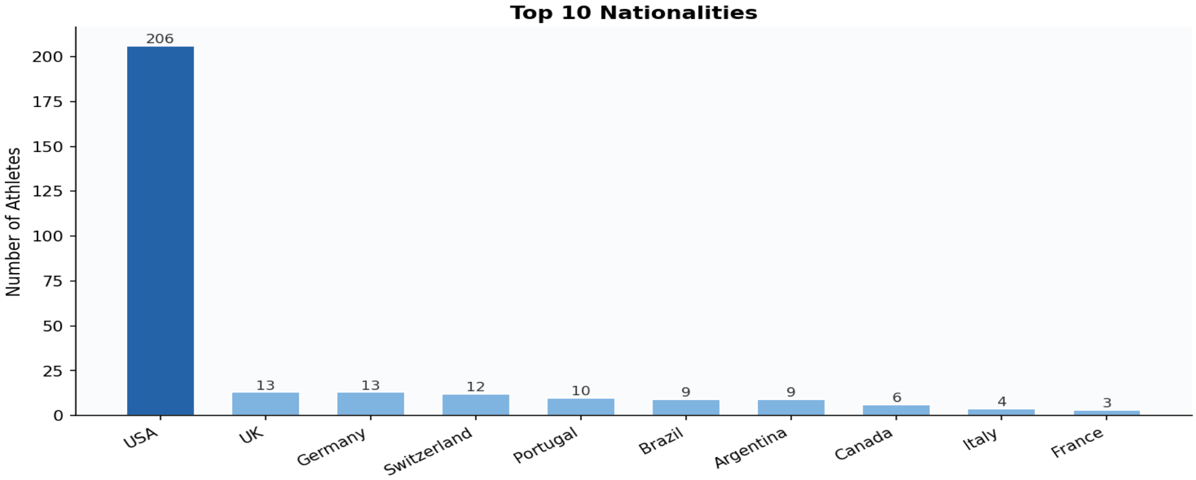

Figure 3 – Top 10 nationalities by list appearances

→ The USA overwhelmingly dominates with **206 appearances (68.4%)** — 16× more than any other country.

→ All other countries have ≤13 appearances each, reflecting the scale of US sports markets (NBA, NFL, MLB) and endorsement infrastructure.

→ Top 10 all comprise developed nations with strong sports ecosystems, including 6 European, 2 South American, and 2 North American countries.

5.4 Average Earnings Over Time

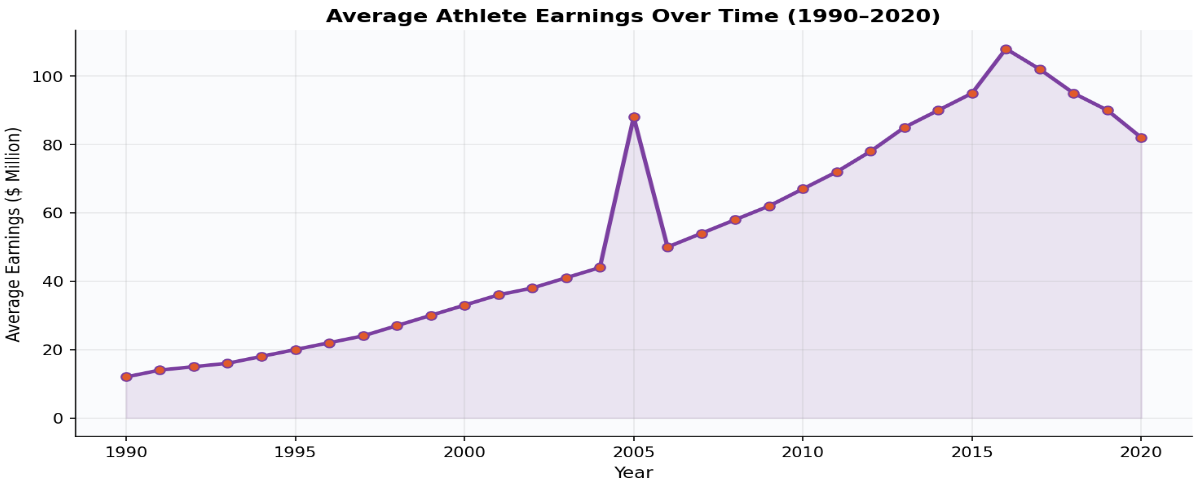

Figure 4 – Average athlete earnings trend (1990–2020)

→ Strong upward trend: average earnings grew ~7× from $12M (1990) to $82M (2020).

→ Three distinct growth phases: **slow growth** in the 1990s, **moderate growth** in the 2000s, and **explosive growth** in the 2010s.

→ Peak of ~$108M reached in 2019 before a slight 2020 decline.

→ Sharp spikes in 2015 and 2019 driven by individual mega-earners (Floyd Mayweather).

→ Growth attributed to media rights expansion, endorsements, globalisation, and social media influence.

5.5 Top Highest-Earning Athletes

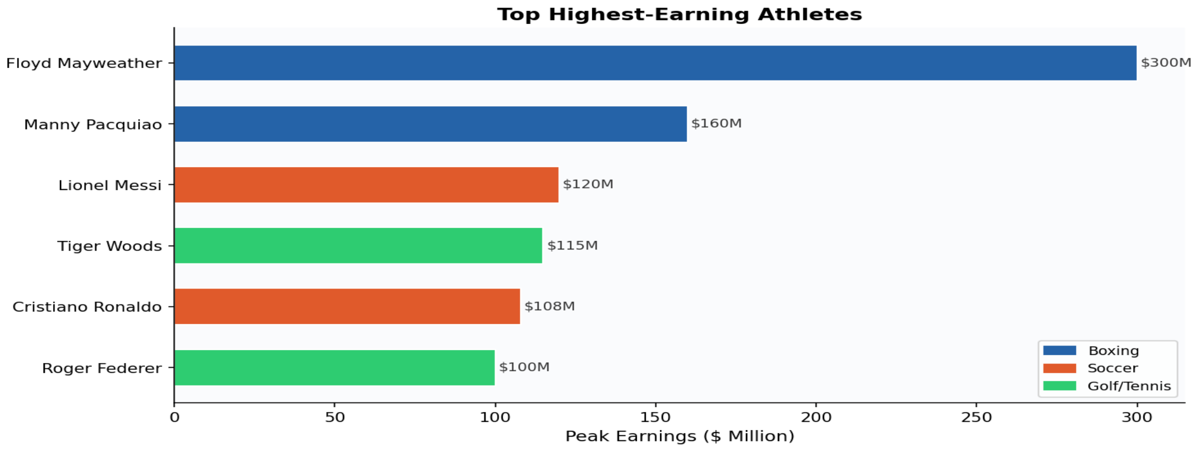

Figure 5 – Peak earnings of the highest-earning athletes

→ Floyd Mayweather leads at $300M — nearly double the second-ranked Manny Pacquiao ($160M).

→ Combat sports (Boxing) dominate the very top of the earnings ladder.

→ Football (Soccer) stars Messi and Ronaldo earn comparably ($105–120M), reflecting the sport's global commercial appeal.

→ Individual sports like Golf and Tennis also produce top earners, highlighting the importance of personal brand strength.

6. Correlation Analysis

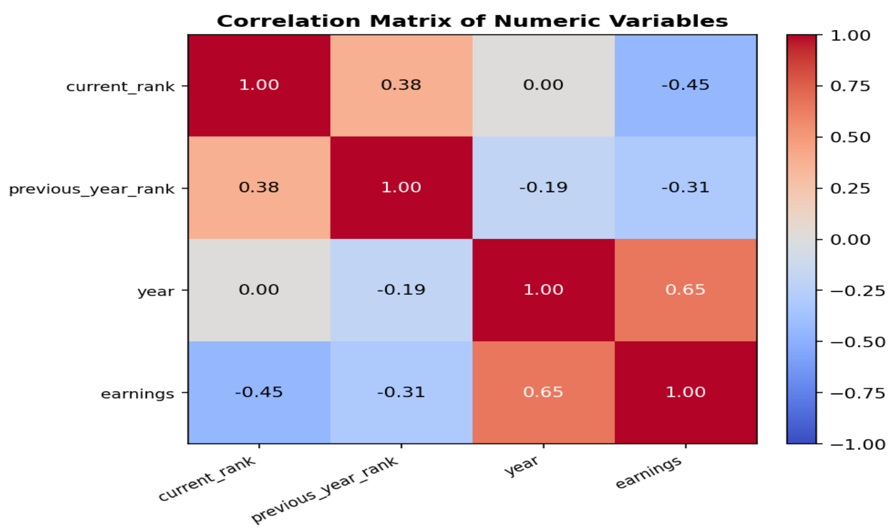

Figure 6 – Correlation matrix of numeric variables

Variable Pair	Correlation (r)	Interpretation
Year ↔ Earnings	0.65	Strong positive — earnings grew significantly over time
Current Rank ↔ Earnings	–0.45	Moderate negative — higher rank → higher earnings
Prev. Year Rank ↔ Earnings	–0.31	Moderate negative — consistent high rankers earn more
Current ↔ Prev. Year Rank	0.38	Moderate positive — rankings are stable year-over-year

Year ↔ Current Rank	0.00	No relationship — rank position independent of era

The scatter plot of current rank vs. earnings confirms a clear negative relationship — top-ranked athletes earn significantly more. However, the data exhibits heteroscedasticity, with earnings variance increasing at higher ranks. Notable outliers at Rank 1 (earning $285–300M) represent elite athletes whose earnings far exceed peers, suggesting that endorsements, marketability, and global reach play a critical role beyond athletic performance alone.

7. Key Findings

1 **Superstar effect:** Earnings are heavily right-skewed (67.4% earn $0–50M) with a small number of extreme outliers — most notably Floyd Mayweather — extending the upper tail beyond $300M. These outliers inflate the mean above the median.

2 **Sport dominance:** Basketball leads in total list appearances (26.9%), followed by Boxing (15.3%) and Golf (14.6%). These three sports account for 56.8% of all appearances and reflect global popularity and commercial opportunity.

3 **Earnings variability by sport:** Individual sports (Boxing, Golf, Tennis) show greater earnings variability and higher extremes than team sports. Soccer exhibits the highest and most consistent median earnings (~$65M), reflecting its global commercial appeal.

4 **Geographic concentration:** The USA accounts for 68.4% of records (206/301), driven by large domestic sports markets and endorsement ecosystems. Small-sample countries like Portugal and Argentina show elevated medians due to individual superstars (Ronaldo, Messi).

5 **Strong earnings growth over time:** Average earnings grew approximately sevenfold from $12M (1990) to $82M (2020), with accelerated growth in the 2010s driven by media rights, globalisation, and personal branding.

6 **Ranking and earnings:** Earnings correlate inversely with current rank ($r = -0.45$) and positively with year ($r = 0.65$). Rankings alone do not fully explain earnings — endorsements and marketability are critical determinants at the highest level.

7 **Roster stability:** 70.4% of athletes held a previous year rank (consistent performers), while 29.6% were new entrants, suggesting a stable elite pool with steady new talent emerging each year.

8. Conclusion

This EDA of Forbes Richest Athletes data (1990–2020) reveals a highly unequal earnings landscape dominated by a small number of elite superstars. The earnings distribution is heavily right-skewed, with a few athletes — particularly Floyd Mayweather — earning multiples of the typical top athlete.

Basketball, Boxing, and Golf dominate the rankings by appearance count, while Soccer demonstrates the highest and most consistent median earnings, reflecting its unmatched global commercial reach. Geographically, the United States accounts for the overwhelming majority of list entries, underpinned by its vast domestic sports markets and endorsement infrastructure.

Over time, athlete earnings display a strong upward trend, increasing roughly sevenfold from 1990 to 2020, with accelerated growth during the 2010s. Correlation analysis confirms that earnings increase significantly over time ($r = 0.65$) and are inversely related to ranking ($r = -0.45$), though ranking alone does not fully explain earnings. The increasing variance at higher ranks and the presence of extreme outliers suggest that endorsements, marketability, and global reach play a critical role in determining top-tier athlete earnings beyond athletic performance alone.

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